GATE

CS & IT

Data Structure & Programming

DPP: 1

Array

Q1 Consider a 1D array a with 9 elements. If the base address of the array is 108 and the size of each array element is 4 bytes, the address of a[7] is- _____
(Assume array index starts from 0)

Q2 Consider a 1D array a[-127.....,+255] where -127 and +255 are the starting index and ending index of the array respectively. The number of elements in the array is _____.

Q3 Consider a 1D array a with 19 elements. If the base address of the array is 108 and the size of each array element is 4 bytes, the address of a[7] is-
(Assume array index starts from -11)_____.

Q4 Consider a 2D array a[-127 to +255][-13 to +14]. The number of elements in the array is _____.

Q5 Consider the natural numbers from 1 to 256 are stored in a 2D array arr[-28 to 3][-3 to 3]. Find the element present at location arr[-16][1]. (Suppose, the elements are stored in row-major order)_____.

Q6 Consider the natural numbers from 1 to 256 are stored in a 2D array arr[-28 to 3][-3 to 3]. Find the address of the location arr[-16][1] if the starting address of the array is 625 and size of each element is 4 bytes. (Suppose, the elements are stored in row-major order)_____.

Q7 Consider the whole numbers from 0 to 127 are stored in a 2D array arr[0 to 15][0 to 7]. Find the element present at location arr[6][4]. (Suppose, the elements are stored in column-major order)_____

Q8 Consider a 2D array arr[-15 to 15][-7 to 7]. Find the address of the location arr[-1][5] if the starting address of the array is 500 and size of each element is 4 bytes. (Suppose, the elements are stored in column-major order) _____

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Answer Key

Q1	136~136	Q5	89~88
Q2	383~383	Q6	977~977
Q3	180~180	Q7	70~66
Q4	10724~10722	Q8	2044~2044

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Hints & Solutions

Q1 Text Solution:

Address of a[7] = 108+(7-0)*4 = 136.

Q2 Text Solution:

Number of elements in the array=255-(-127)+1=383

Q3 Text Solution:

Address of a[7] = 108+(7-(-11))*4 = 180.

Q4 Text Solution:

Number of elements in each row= 255-(-127)+1=383

Number of elements in each column = +14-(-13)+1=28

Number of elements in the 2D array = 383*28 = 10724

Q5 Text Solution:

Number of elements in each column=3-(-3)+1=7

The element present at arr[-16][1]

=(-16-(-28))*7+(1-(-3)+1)

=89.

Q6 Text Solution:

Number of elements in each column= 3-(-3)+1=7

Address of location arr[-16][1]

= 625 + [(-16-(-28))*7+(1-(-3))]*4

= 977

Q7 Text Solution:

Number of elements in each row= 16

The element present at arr[6][4]=(4-0)*16+(6-0)=70

Q8 Text Solution:

Number of elements in each column= 7-(-7)+1=15

Address of location arr[-1][5]

= 500 + [(5-(-7))*(15-(-15)+1)+(-1-(-15))]*4

= 500 + [12 *31 +14] *4

= 500 + [372 + 14]*4

= 2044

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